

IMAGE PROCESSING PIPELINES FOR OFFLINE AND ONLINE 3D RECONSTRUCTION WITH A MONOCULAR USB CAMERA

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ABSTRACT

We propose image preprocessing pipelines for 3D reconstruction using affordable 640×480 USB cameras. Our approach addresses photometric and geometric inconsistencies that degrade reconstruction quality in commodity hardware. We present two complementary pipelines: an online version optimized for real-time direct sparse odometry (DSO-SLAM) [1] and an offline version tailored for feature-based structure-from-motion (COLMAP) [2].

The online pipeline integrates gamma correction, fixed-gain exposure compensation, bilateral filtering, and integrated photometric-geometric undistortion, prioritizing photometric consistency for direct methods. The offline pipeline extends these steps with a novel tuned Contrast Limited Adaptive Histogram Equalization (CLAHE) [3] method and unsharp masking to maximize feature detectability.

Evaluation on diverse indoor and outdoor datasets demonstrates significant improvements: the online pipeline achieves 35.8% improvement in tracking robustness and 10.9% improvement in temporal consistency, while the offline pipeline reduces reprojection errors by up to 9.1% and increases point cloud density by $2.2\times$. These findings suggest that for photogrammetric applications, algorithmic feature density and photometric consistency should take precedence over human perceptual quality metrics. Both versions show substantial improvement over unprocessed camera input, validating the importance of targeted preprocessing for robust 3D reconstruction with commodity hardware. The source code and supplementary materials are available at our project page.

Index Terms— Image preprocessing, Photometric calibration, 3D reconstruction, Direct sparse odometry, Structure-from-motion, Commodity cameras

1. INTRODUCTION

Three-dimensional reconstruction from images has emerged as a fundamental capability in computer vision, enabling applications ranging from autonomous navigation and augmented reality to digital preservation and virtual tourism. Current state-of-the-art reconstruction methods, whether

feature-based approaches like COLMAP [2] or direct methods like DSO-SLAM [1], heavily rely on high-quality camera hardware with precise calibration, low noise, and consistent photometric properties. However, this dependency becomes a significant hindrance when hardware resources are limited, particularly in scenarios involving commodity USB cameras, embedded systems, or budget-constrained applications.

The challenge is particularly pronounced with low-resolution cameras operating at 640×480 resolution, which are ubiquitous in consumer electronics but exhibit substantial photometric and geometric inconsistencies. These cameras typically apply non-linear tone curves, automatic exposure and gain control, white balance adjustments, and suffer from lens distortion and vignetting, all optimized for human visual perception rather than geometric reconstruction. When such images are directly fed into 3D reconstruction pipelines, the fundamental assumptions underlying both feature-based and direct methods are violated: feature-based approaches struggle with inconsistent feature detection and matching, while direct methods suffer from biased photometric residuals that lead to tracking failures and reconstruction drift.

To address this fundamental limitation, we propose a comprehensive image preprocessing framework that bridges the gap between commodity camera output and the requirements of high-quality 3D reconstruction. Our approach recognizes that different reconstruction paradigms have distinct requirements: direct methods like DSO-SLAM prioritize photometric consistency across frames, while feature-based methods like COLMAP benefit from enhanced feature detectability and matching robustness. Consequently, we design two complementary preprocessing pipelines: one optimized for real-time online reconstruction and another tailored for offline structure-from-motion; both of which are operating on the same affordable 640×480 camera hardware.

The online pipeline targets direct sparse odometry applications, where maintaining photometric consistency is paramount. We implement a lightweight preprocessing chain that includes gamma correction, fixed-gain exposure compensation, edge-preserving bilateral filtering, and integrated photometric-geometric undistortion. This pipeline prioritizes temporal photometric stability over visual enhancement, ensuring that corresponding pixels maintain consistent intensities across frames, a critical requirement for direct methods

Thanks to MPAC Labs for providing USB camera.

that minimize photometric error. The entire pipeline runs in real-time at 30 fps, making it suitable for live reconstruction scenarios.

The offline pipeline extends these preprocessing steps with additional enhancement stages specifically designed for feature-based reconstruction. We introduce a novel tuned Contrast Limited Adaptive Histogram Equalization (CLAHE) method with optimized parameters that maximize feature detectability without introducing artifacts, followed by unsharp masking for gradient enhancement. These enhancements are carefully calibrated to improve feature matching robustness while maintaining geometric accuracy, addressing the distinct requirements of structure-from-motion pipelines.

Our key insight is that for photogrammetric applications, algorithmic feature density and photometric consistency should take precedence over human perceptual quality metrics. Traditional image enhancement techniques optimized for visual appeal often introduce artifacts that degrade reconstruction quality, while targeted preprocessing that addresses the specific needs of reconstruction algorithms yields superior results. This principle guides our design decisions throughout both pipelines.

We evaluate our approach on diverse indoor and outdoor datasets, including challenging scenarios with motion-activated sensor lights that cause abrupt illumination changes. Quantitative results demonstrate substantial improvements: the online pipeline achieves a 35.8% improvement in tracking robustness and a 10.9% improvement in temporal consistency compared to raw camera input, while the offline pipeline reduces reprojection errors by up to 9.1% and increases point cloud density by $2.2\times$ compared to baseline methods. These improvements validate that targeted preprocessing can significantly enhance reconstruction quality even with commodity hardware, making high-quality 3D reconstruction accessible without specialized cameras.

Contributions. Our main contributions are:

- A comprehensive preprocessing framework for 3D reconstruction that addresses both photometric and geometric inconsistencies in commodity cameras, designed to work with affordable 640×480 hardware.
- Two complementary pipelines: an online version optimized for real-time direct sparse odometry with emphasis on photometric consistency, and an offline version tailored for feature-based structure-from-motion with enhanced feature detectability.
- A novel tuned CLAHE method with optimized parameters specifically designed to maximize feature matching robustness in structure-from-motion applications.
- Extensive evaluation demonstrating significant improvements in both online and offline reconstruction

scenarios, with quantitative analysis showing substantial gains in tracking robustness, temporal consistency, reprojection accuracy, and point cloud density.

- The insight that algorithmic feature density and photometric consistency should take precedence over human perceptual quality metrics in photogrammetric applications.

Paper Organization. The remainder of this paper is organized as follows. Section 2 reviews related work in image preprocessing for 3D reconstruction. Section 3.1 provides an overview of our preprocessing pipelines, while Section 3.3 describes the offline calibration procedures. Section 3.2 and 3.4 details the online preprocessing pipeline, and Section ?? further illustrates our offline pipeline. For experimental results and analysis, we discuss thoroughly in Section 4 with additional ablation study in Section 4.5. Finally, Section ?? concludes with a discussion of limitations and future directions.

2. RELATED WORK

Visual 3D reconstruction and camera pose estimation are fundamental problems in computer vision. The literature is broadly categorized into two paradigms based on processing constraints and data ingestion: offline Structure from Motion (SfM), which prioritizes global accuracy using unordered image collections, and online Simultaneous Localization and Mapping (SLAM), which emphasizes real-time tracking from sequential video streams. In this work, we specifically leverage COLMAP [2] as the representative framework for offline reconstruction and Direct Sparse Odometry (DSO) [1] for online processing.

2.1. Structure from Motion

Structure from Motion (SfM) is the standard approach for recovering sparse 3D structure and camera poses from a set of images. Among modern SfM pipelines, COLMAP [2] is widely regarded as the state-of-the-art for its robustness and accuracy. The general pipeline, adopted by COLMAP, consists of four key stages:

1. **Feature Extraction and Matching:** The process begins by extracting varying-invariant features, typically SIFT [4], from all input images. These features are matched across image pairs and subjected to rigorous geometric verification to filter out outliers and establish reliable epipolar geometry.
2. **Initialization:** The reconstruction is initialized by selecting a seed image pair with sufficient baseline and feature correspondences. The relative pose is recovered

using the Essential matrix, and common feature points are triangulated to form an initial 3D cloud.

3. **Incremental Registration:** New images are sequentially registered to the existing model using the Perspective- n -Point (PnP) algorithm. COLMAP employs a robust next-best-view selection strategy to maximize stability during this expansion.
4. **Bundle Adjustment (BA):** To mitigate drift accumulation, a global non-linear optimization is performed. BA jointly refines the 3D structure points $\mathbf{X}_j$ and camera parameters $\mathbf{C}_i$ by minimizing the total reprojection error:

$$E_{BA} = \sum_{i,j} \rho(\|\mathbf{p}_{i,j} - \Pi(\mathbf{C}_i, \mathbf{X}_j)\|_{\Sigma}^2)$$

where Π denotes the projection function, $\mathbf{p}_{i,j}$ is the observed feature location, and ρ is a robust loss function.

Due to its batch nature and exhaustive geometric verification, COLMAP serves as an ideal baseline for generating high-fidelity ground truth in offline scenarios.

2.2. Simultaneous Localization and Mapping

In contrast to SfM, Visual SLAM targets real-time pose estimation and mapping. SLAM approaches are generally divided into *indirect* (feature-based) methods, such as ORB-SLAM [5], and *direct* methods, such as DSO [1].

While indirect methods abstract images into feature descriptors [6], minimizing geometric reprojection error, Direct Sparse Odometry (DSO) operates directly on pixel intensities. It omits the feature extraction step and instead optimizes the photometric error over a window of active keyframes. The objective function in DSO minimizes the intensity difference between a point in a reference frame I_i and its projection in a target frame I_j :

$$E_{photo} = \sum_{p \in \Omega} w_p \|I_j(p') - I_i(p)\|_{\gamma}$$

where p' is the projected pixel location of point p , and $\|\cdot\|_{\gamma}$ denotes the Huber norm.

Because direct methods like DSO rely on the brightness constancy assumption that a 3D point appears with the same intensity across different views; they are theoretically more precise than feature-based methods in texture-poor environments. However, this also makes them highly sensitive to photometric calibration errors, such as vignetting, non-linear response functions (gamma), and auto-exposure changes. This vulnerability underscores the necessity of the strictly calibrated image processing pipeline proposed in this paper.

3. PROPOSED METHOD

3.1. Overview

In this work, we use a direct visual SLAM approach, specifically, Direct Sparse Odometry (DSO), to perform real-time, online 3D reconstruction using a monocular USB camera. Although direct methods offer high accuracy and efficiency, their performance is highly dependent on the quality and consistency of the input images, which introduces several practical challenges.

Firstly, commodity USB cameras are primarily designed for human visualization rather than geometric reconstruction. Their output images typically undergo undocumented internal processing, including nonlinear tone mapping, automatic exposure control, and inhomogeneous brightness attenuation caused by lens vignetting. Additionally, lens distortion is common and violates the pinhole camera model assumed by most SLAM systems. These factors introduce significant photometric and geometric inconsistencies across frames, which are especially problematic for direct methods.

Secondly, the entire system must operate within real-time constraints. At a resolution of 640×480 and 30 frames per second, the image processing pipeline must be computationally lightweight and avoid expensive per-frame optimization or iterative procedures.

The DSO formulation of visual odometry is based on minimizing photometric error between corresponding pixels across frames. Unlike feature-based methods, it uses pixel intensities and their spatial gradients directly for motion estimation. Consequently, DSO relies on the strong assumption of brightness constancy: the intensity of a scene point should remain consistent across frames, except for global affine changes. In practice, violations of this assumption, such as non-linear camera response, exposure fluctuations, and spatial brightness falloff, lead to biased residuals and reduced tracking robustness. Furthermore, high-gradient regions such as edges and corners play a critical role in the optimization, making local contrast preservation particularly important.

Motivated by these observations, we have designed a carefully ordered image pre-processing pipeline that explicitly targets the dominant sources of photometric and geometric inconsistency in USB camera images, while maintaining real-time performance. The pipeline combines offline calibration and online processing. Offline calibration is used to estimate camera intrinsics, lens distortion, photometric response, and vignetting characteristics. During online operation, each incoming frame is processed through a sequence of fast, deterministic operations, including photometric linearisation, exposure stabilisation, vignetting correction, geometric undistortion, grayscale conversion, and light denoising, to produce images that are better suited to direct photometric alignment in DSO.

The design philosophy is not to enhance image appearance aggressively, but to restore a representation that more

closely matches the assumptions of direct visual SLAM. By prioritising photometric consistency, gradient preservation, and computational efficiency, the proposed pipeline significantly improves the suitability of raw USB camera frames for real-time, direct 3D reconstruction.

3.2. Online Pipeline

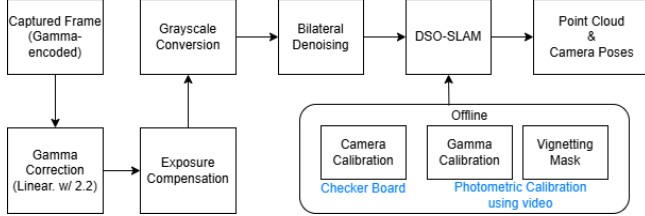


Fig. 1. Our Online Reconstruction Pipeline

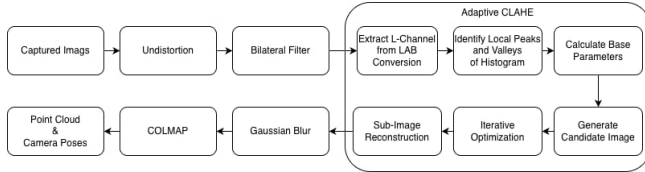


Fig. 2. Offline Calibration Pipeline

Our target application is a direct sparse odometry (DSO) system running on frames captured from a commodity USB camera at a resolution of 640×480 and 30 fps. In contrast to feature-based SLAM, direct methods such as DSO directly minimize photometric error between image intensities. As a result, their performance strongly depends on both *geometric* and *photometric* consistency across frames.

However, off-the-shelf USB cameras are typically optimized for human viewing rather than geometric reconstruction. They apply non-linear tone curves, automatic exposure, white balance, and suffer from lens distortion and vignetting. If these effects are not compensated, the basic brightness constancy assumption underlying direct methods is violated, leading to biased residuals, slower convergence, and tracking failures.

To address this, we design a lightweight pre-processing pipeline that runs before DSO and explicitly targets the following goals:

- **Geometric consistency:** correct lens distortion so that the camera can be well approximated by a pinhole model using calibrated intrinsics.
- **Photometric consistency:** reduce non-linear response, exposure fluctuations, and vignetting so that corresponding pixels have comparable intensities across frames.

- **Real-time efficiency:** keep the per-frame cost low enough to maintain real-time processing at 30 fps on 640×480 images.

Driven by these objectives, we propose a two-stage processing chain applied to each incoming USB camera frame. The first stage consists of software-based preprocessing steps implemented in our custom `PipelineProcessor` class, while the second stage integrates photometric and geometric undistortion within DSO’s `Undistort` framework. The complete pipeline, in order of application, is:

1. Gamma correction using a fixed power curve ($\gamma = 2.2$);
2. Fixed gain exposure compensation computed from the first frame;
3. Grayscale conversion using BT.709 luminance weights;
4. Light bilateral filtering for edge-preserving denoising;
5. Photometric undistortion using photometric calibration (or inverse camera response function (CRF)) and vignetting correction;
6. Geometric undistortion using calibrated intrinsics and distortion coefficients.

All steps are implemented using efficient per-pixel operations or lookup tables in OpenCV [7], which keeps the overall overhead small while improving the suitability of USB camera images for direct 3D reconstruction.

3.3. Offline Camera and Photometric Calibration

Before running the online pipeline, we perform an offline calibration step to estimate the geometric and photometric parameters of the USB camera. The calibration results are stored in separate files and reused at runtime.

Geometric calibration. We estimate the intrinsic camera matrix $\mathbf{K}$ and distortion coefficients $\mathbf{d}$ using standard checkerboard calibration in OpenCV [7] with the rational distortion model. The intrinsics follow the pinhole model

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

and the vector $\mathbf{d} = (k_{1..6}, p_{1,2})$ accounts for radial and tangential distortion. These are stored to generate an undistortion map for efficient rectification. The calibration board we utilized is shown in 3(a).

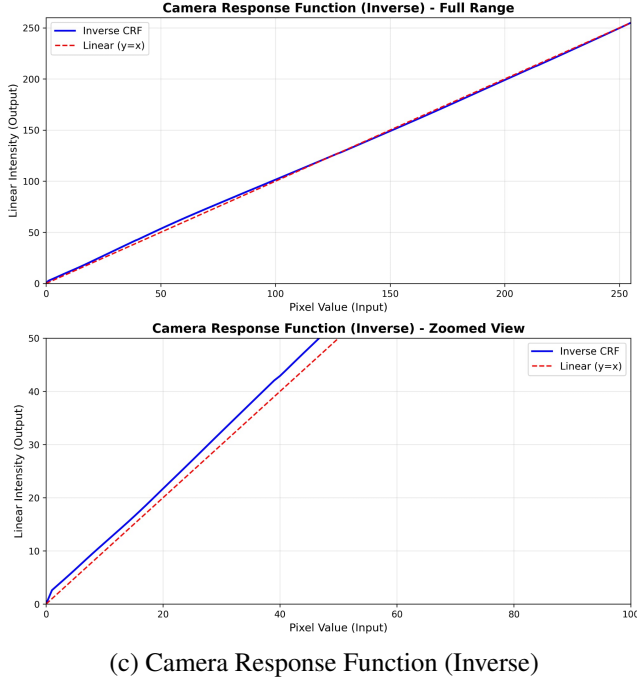
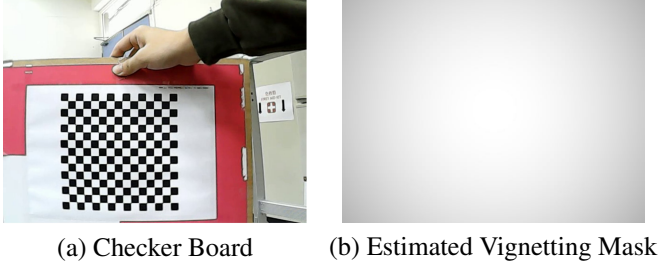


Fig. 3. Calibration board captured by our USB camera with the effect of barrel distortion (left), and the estimated photometric calibration result using [8] (right). (c) show the inverse camera response function

Photometric calibration. We employ the online photometric calibration method [8] to jointly estimate the camera response function (CRF) and vignetting parameters from a video sequence of corridor scenes with wide range of light variation. The estimated vignetting mask and photometric calibration curve are shown in 3(b) and 3(c), respectively. The calibration process analyzes brightness variations across frames and spatial locations to recover both the inverse response function and the radial vignetting model.

The inverse camera response function is stored as a 256-entry lookup table in `pcalib.txt`, where each entry maps an 8-bit pixel value to its corresponding linear intensity. The vignetting is modeled as a polynomial function of the normalized radial distance from the image center:

$$V(r) = 1 + v_1 r^2 + v_2 r^4 + v_3 r^6, \quad (2)$$

where $r \in [0, 1]$ is the normalized distance from the principal

point, and v_1, v_2, v_3 are the estimated vignetting coefficients. The vignetting mask is stored as a 16-bit grayscale image (`vignette.png`), where pixel values represent the attenuation factor at each image location, normalized such that the center value is 65535 (maximum brightness).

This approach provides more accurate photometric calibration than manual white sheet capture, as it leverages natural scene statistics and automatically accounts for exposure variations during calibration.

3.4. Online Pre-processing Pipeline

Leveraging the offline calibration, each incoming frame undergoes a lightweight, real-time sequence of photometric linearization followed by geometric undistortion. This ensures compliance with DSO’s pinhole and brightness constancy assumptions.

3.4.1. Gamma Correction

Consumer USB cameras typically output images in a gamma-compressed color space similar to sRGB. As direct methods assume a roughly linear relationship between scene irradiance and pixel intensities, it is beneficial to perform a partial linearization step. We implement gamma correction using a per-pixel power operation on normalized intensity values in RGB color space.

Given an input pixel value $I \in [0, 255]$ for each color channel, the corrected value is obtained as

$$I' = \left(\frac{I}{255} \right)^\gamma \cdot 255, \quad (3)$$

where $\gamma = 2.2$ is the standard gamma value for sRGB linearization. The operation is implemented using `cv::pow` on floating-point normalized values, followed by conversion back to 8-bit integers per channel.

This fixed gamma correction serves as an approximate inversion of the camera response function and effectively reduces strong non-linearities in the RGB color space. Note that a more accurate camera response function (CRF) calibration is applied later in the photometric undistortion stage using the `pcalib.txt` lookup table; however, the fixed gamma correction provides a reliable preprocessing step that improves the linearity of the image data before subsequent processing stages.

3.4.2. Fixed Gain Exposure Compensation

Direct sparse odometry assumes that corresponding scene points exhibit approximately consistent intensities across frames, up to a global affine brightness change. Commodity USB cameras, however, often apply automatic exposure and gain, causing inter-frame brightness variation that can bias photometric residuals. To address this, we employ a *fixed*

gain exposure compensation strategy that maintains photometric consistency throughout the sequence.

Fixed Gain Strategy. Rather than adapting the gain dynamically per frame, we compute a single global scaling factor from the first frame and apply it uniformly to all subsequent frames. This approach prioritizes photometric consistency over adaptive brightness adjustment, which is crucial for direct methods like DSO that rely on stable photometric residuals.

Let μ_0 denote the mean grayscale intensity of the first frame, computed using BT.709 luminance weights. We compute a target gain g_{fixed} that maps μ_0 to a mid-gray reference value (typically 128):

$$g_{\text{fixed}} = \frac{I_{\text{target}}}{\mu_0 + \epsilon}, \quad (4)$$

where ϵ is a small constant to avoid division by zero, and I_{target} is the desired mean intensity. The gain is then clamped to a safe range $[g_{\text{min}}, g_{\text{max}}]$ to prevent extreme amplification of noise or saturation. Once computed, this fixed gain is applied to all frames in the sequence:

$$I_{\text{compensated}}(x, y) = \text{clip}(g_{\text{fixed}} \cdot I(x, y), 0, 255), \quad (5)$$

where the clipping is performed per RGB channel.

This fixed gain approach ensures that the photometric relationship between frames remains stable, which is more important for DSO than maintaining a specific absolute brightness level. By avoiding per-frame adaptation, we eliminate temporal brightness fluctuations that could introduce spurious photometric errors in the direct alignment process.

3.4.3. Grayscale Conversion

DSO operates on single-channel intensity images and uses image gradients for direct photometric alignment. After completing all color-space and photometric corrections in RGB space, we convert the image to grayscale using the BT.709 luminance definition, which better reflects human and sensor sensitivity to different color channels than a simple average.

Given a color pixel (R, G, B) in the range $[0, 255]$, the corresponding luminance Y is computed as

$$Y = 0.2126 R + 0.7152 G + 0.0722 B. \quad (6)$$

We implement this conversion explicitly using per-channel weighted summation rather than relying on the OpenCV default, to ensure consistency between the calibration step and the runtime pipeline. The result is clipped to $[0, 255]$ and stored as an 8-bit grayscale image.

Working in a single luminance channel reduces the computational cost of the subsequent SLAM pipeline and simplifies the photometric model, while preserving the intensity gradients that are most relevant for direct alignment.

3.4.4. Bilateral Filter Denoising

To mitigate high-frequency sensor noise without degrading the intensity gradients essential for direct tracking, we employ a lightweight bilateral filter. Unlike standard Gaussian smoothing, this approach preserves structural edges by weighting neighbors based on both spatial proximity and intensity similarity. We empirically utilize a kernel diameter $d = 5$ with spatial and color standard deviations $\sigma_s = \sigma_c = 20$. This configuration effectively stabilizes the photometric error surface for DSO optimization while maintaining the sharpness of fine textures required for accurate pose estimation.

Empirically, we found that this mild bilateral filtering is sufficient to stabilize the intensity measurements for DSO without noticeably degrading the sharpness of the image or the quality of the reconstructed geometry. The edge-preserving property is particularly important for direct methods, as it maintains the gradient structure that DSO relies on for photometric alignment.

3.4.5. Photometric and Geometric Undistortion

After the software-based preprocessing steps described above, the grayscale image undergoes integrated photometric and geometric undistortion, which is implemented within DSO’s `Undistort` class. This stage combines camera response function (CRF) correction, vignetting compensation, and lens distortion rectification in a unified framework.

Photometric Calibration. The photometric undistortion applies the inverse camera response function (CRF) and vignetting correction using calibration data obtained from offline photometric calibration. The CRF is stored as a 256-entry lookup table in `pcalib.txt`, where each entry maps an 8-bit pixel value to its corresponding linear intensity. The vignetting mask is stored as a 16-bit grayscale image (`vignette.png`), where pixel values represent the attenuation factor at each image location, normalized such that the center value is 65535.

The photometric correction is applied per-pixel as:

$$I_{\text{photometric}}(x, y) = \frac{\text{CRF}^{-1}(I(x, y))}{V(x, y)}, \quad (7)$$

where CRF^{-1} is the inverse response function lookup and $V(x, y)$ is the vignetting mask value at location (x, y) , normalized to the range $(0, 1]$ with $V(c_x, c_y) \approx 1$ near the principal point. The division by the vignetting mask amplifies darker corner regions to match the center brightness, effectively compensating for radial light falloff.

Geometric Undistortion. Using the intrinsic matrix $\mathbf{K}$ and distortion coefficients $\mathbf{d}$ obtained during offline geometric calibration, we precompute an undistortion map that rectifies

radial and tangential lens distortion. The undistortion map consists of two lookup tables map_1 and map_2 that, for each pixel in the undistorted image, specify the corresponding sampling location in the original distorted image.

At runtime, geometric undistortion is applied using bilinear interpolation:

$$I_{\text{undist}}(x, y) = \text{interp}(I_{\text{photometric}}, \text{map}_1(x, y), \text{map}_2(x, y)), \quad (8)$$

where interp denotes bilinear interpolation. The maps are computed once during initialization using the current image size and then reused for all subsequent frames, ensuring efficient real-time processing.

By rectifying the images in this way, we ensure that the camera model used by DSO is consistent with the actual pixel geometry, which improves the accuracy of both pose estimation and depth reconstruction. The integrated approach of applying photometric and geometric corrections together ensures that all corrections are performed in the correct order and that the final image is optimally prepared for direct sparse odometry.

3.4.6. Pipeline Summary

The complete processing pipeline, in order of application, is:

1. **Gamma correction:** Linearize gamma-compressed RGB image using fixed power curve ($\gamma = 2.2$)
2. **Fixed gain exposure compensation:** Apply uniform scaling factor computed from first frame to RGB channels
3. **Grayscale conversion:** Convert to single-channel using BT.709 luminance weights
4. **Bilateral filter denoising:** Light edge-preserving noise reduction on grayscale image
5. **Photometric undistortion:** Apply CRF inverse lookup and vignetting correction using `pcalib.txt` and `vignette.png`
6. **Geometric undistortion:** Rectify lens distortion using precomputed remap tables from `camera.txt`

All steps are designed to maintain photometric consistency while preserving geometric structure, which is essential for reliable direct sparse odometry. The pipeline is implemented to run in real-time at 30 fps on commodity hardware, with the majority of computational cost in the bilateral filter and geometric remapping operations, both of which are highly optimized in OpenCV. The software-based preprocessing stages (gamma correction, exposure compensation, grayscale conversion, and bilateral filtering) are implemented in our custom `PipelineProcessor` class, while the photometric and geometric undistortion stages are integrated within DSO’s existing `Undistort` framework, ensuring seamless compatibility with the SLAM system.

3.5. Offline Pre-processing Pipeline

Unlike the online pipeline where computational latency is the primary constraint, the offline pipeline prioritizes maximum feature recoverability for Structure-from-Motion (SfM). We propose a sequential enhancement strategy designed to maximize the number of valid SIFT keypoints in texture-poor environments. The pipeline consists of three stages: edge-preserving smoothing, adaptive contrast enhancement, and high-frequency boosting.

3.5.1. Adaptive Bilateral Filtering

To mitigate sensor noise without compromising the high-frequency gradients essential for feature detection, we employ a bilateral filter [9]. Unlike standard Gaussian smoothing which averages pixels solely based on spatial distance, the bilateral filter incorporates a photometric term to preserve structural edges. The filtered intensity $I_{bf}(\mathbf{x})$ at pixel coordinate $\mathbf{x}$ is defined as:

$$I_{bf}(\mathbf{x}) = \frac{1}{W_p} \sum_{\mathbf{x}_i \in \Omega} I(\mathbf{x}_i) G_{\sigma_s}(\|\mathbf{x} - \mathbf{x}_i\|) G_{\sigma_r}(|I(\mathbf{x}) - I(\mathbf{x}_i)|) \quad (9)$$

where Ω is the spatial neighborhood, and G_{σ_s} , G_{σ_r} differ as spatial and range Gaussian kernels, respectively. We empirically utilize a kernel diameter $d = 9$ with standard deviations $\sigma_s = \sigma_r = 75$. This specific configuration effectively suppresses noise in homogeneous regions (e.g., walls) while retaining sharp intensity transitions at object boundaries, preventing the “defocusing” effect common in linear filtering.

3.5.2. Tuned-CLAHE with Histogram Analysis

Standard CLAHE [3] relies on fixed parameters (e.g., a clip limit of 2.0), which often leads to noise amplification in flat regions or insufficient enhancement in dark areas. We propose a *Tuned-CLAHE* approach that adaptively estimates optimal parameters based on the image’s histogram topology. The algorithm proceeds in three phases:

1. Histogram Analysis and Parameter Derivation: We first analyze the histogram to identify local peaks and valleys. To determine a data-driven clip limit, we calculate the N -parameter (average valley height):

$$N_{\text{param}} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \text{freq}(v) \quad (10)$$

where $\mathcal{V}$ is the set of identified valleys. The Adaptive Clip Limit (I_{CL}) is then derived relative to the peak frequencies:

$$I_{CL} = \frac{\text{freq}(p_i) + N_{\text{param}}}{\max_{v \in \mathcal{V}}(\text{freq}(v))} \quad (11)$$

Simultaneously, the grid size (Contextual Block Dimension, CBD) is scaled logarithmically with the number of

histogram peaks N_{peaks} to match the scene’s complexity: $CBD = \lceil \ln(N_{peaks}) \rceil$.

2. Optimization via Image Quality Metric: Since multiple candidate parameters may be generated, we select the optimal configuration by maximizing a Comprehensive Image Quality Index (CIQI). The CIQI balances contrast gain against noise and brightness distortion:

$$CIQI = \frac{PSNR + \mathcal{H}}{AMBE} \quad (12)$$

where $\mathcal{H}$ denotes entropy (information richness), PSNR measures signal fidelity, and AMBE (Absolute Mean Brightness Error) penalizes excessive global brightness shifts.

3. Sub-Image Decomposition: To handle non-uniform illumination (e.g., a bright window next to a dark corridor), the image is vertically partitioned into ONS sub-images, where $ONS = \lceil \log_2(|\mathcal{V}|) \rceil$ (adjusted to be even). The optimized CLAHE is applied independently to each strip, ensuring local contrast is maximized without over-saturating already bright regions.

3.5.3. Unsharp Masking for Feature Boosting

Following contrast enhancement, we apply unsharp masking to amplify high-frequency components, thereby increasing the distinctiveness of corner features for the SIFT detector. The sharpened image I_{sharp} is computed as a weighted difference between the source and its low-pass filtered version:

$$I_{sharp}(\mathbf{x}) = I_{clahe}(\mathbf{x}) + \lambda (I_{clahe}(\mathbf{x}) - (G_\sigma * I_{clahe})(\mathbf{x})) \quad (13)$$

where $*$ denotes convolution with a Gaussian kernel. We implement this efficiently as a linear combination $I_{sharp} = (1 + \lambda)I_{clahe} - \lambda I_{blur}$. This step recovers fine texture details that may have been compressed during tone mapping, directly contributing to the increased number of keypoints observed in our experiments.

4. EXPERIMENTS AND DISCUSSION

4.1. Data Acquisition

We evaluate our method on four datasets Table 1, comprising one indoor object-centric scene and three outdoor scenes. While all environments are predominantly static, they exhibit distinct lighting challenges. Notably, two of the outdoor sequences were captured in a long corridor featuring motion-activated sensor lights. These conditions introduce severe and abrupt illumination changes, posing a significant challenge for direct methods like DSO, which rely on brightness constancy. In contrast, the third outdoor scene was captured under consistent lighting conditions, serving as a baseline where our proposed pipeline is expected to perform robustly.

Table 1. Datasets Overview

Dataset	Type	Lighting Variations
Psyduck	Indoor	
Corridor _I	Outdoor	
Corridor _L	Outdoor	✓
Corridor _{Loop}	Outdoor	✓

4.2. Implementation Details

All image sequences were acquired using a commodity USB camera. This UVC-compliant device operates at a resolution of 640×480 (0.3 MP) and features built-in automatic gain, exposure, and white balance controls. For the online reconstruction experiments, the camera was interfaced directly with an Apple MacBook Air (M3 processor) via a USB-C adapter, with all subsequent processing executed in the macOS environment. In contrast, the offline reconstruction tasks were performed on a Windows workstation equipped with an NVIDIA GeForce RTX 4060 GPU.

4.3. Experiment Setup

To comprehensively evaluate the effectiveness of our proposed pipeline, we conduct experiments in two distinct operational modes: *online reconstruction* and *offline reconstruction*. This allows us to assess both real-time performance characteristics and the ultimate reconstruction quality achievable when computational constraints are relaxed. We additionally pick direct method for online 3D reconstruction and feature-based method for offline reconstruction to even better to highlight the differences and distinct challenges between them.

4.3.1. Online Reconstruction Configuration

For online reconstruction experiments, we employ a dual-path architecture that simultaneously processes the same input stream through two parallel pipelines: (1) the *Raw*, which applies only photometric calibration (inverse response function from `pcalib.txt`) without geometric undistortion or vignette compensation, and (2) the *Ours*, which applies our complete photometric-stable processing chain. This side-by-side comparison enables direct, frame-synchronized evaluation of the impact of our proposed enhancements.

The online system operates at the camera’s native frame rate (approximately 30 FPS for the USB camera used), with all processing executed in real-time on the host CPU. The dual-path architecture ensures that both reconstructions receive identical input frames, eliminating temporal misalignment as a confounding factor. We configure DSO with identical intrinsic parameters for both paths, with the only difference being the pre-processing applied to the input images. This controlled setup allows us to isolate the contribution of our pipeline enhancements to the final reconstruction quality.

4.3.2. Offline Reconstruction Configuration

For offline reconstruction, we process pre-recorded video sequences through COLMAP [2] without specific order of input images. This enables us to evaluate reconstruction quality on longer sequences and more challenging scenarios where online processing might be constrained by computational resources and not performs robust enough. The offline reconstruction also allows for more comprehensive quantitative analysis, as we can export complete trajectory data, point clouds, and detailed metrics without concerns about real-time performance. We even perform dense reconstruction to provide higher quality reconstruction.

4.3.3. Evaluation Metrics

We employ distinct evaluation strategies for online and offline reconstruction scenarios, reflecting the different objectives and constraints of each type.

Online Reconstruction Metrics: For online reconstruction, we focus on trajectory quality metrics that directly reflect the stability and consistency of real-time visual odometry estimation. These metrics are computed from the exported camera trajectory data and capture the fundamental characteristics that determine SLAM reliability:

- *Translation Smoothness Score:* Measures the consistency of inter-frame translations, computed as $1 - \frac{\sigma_{\text{trans}}}{\mu_{\text{trans}} + \epsilon}$, where σ_{trans} and μ_{trans} are the standard deviation and mean of translation magnitudes, respectively. Higher values (closer to 1.0) indicate smoother, more consistent motion, which is critical for stable tracking in direct methods like DSO.
- *Rotation Smoothness Score:* Analogous to translation smoothness, but computed for rotational motion between consecutive frames. This metric quantifies the stability of orientation estimation, which is particularly important for applications requiring accurate pose recovery.
- *Tracking Robustness:* A stability metric computed as $\frac{1}{1+CV}$, where CV is the coefficient of variation (standard deviation divided by mean) of translation magnitudes. Values closer to 1.0 indicate more stable tracking with less variance, directly reflecting the system's ability to maintain consistent motion estimates under varying illumination conditions.
- *Temporal Consistency:* Measures the stability of rotational motion over time, computed similarly to tracking robustness but for rotation angles. This metric captures the temporal coherence of orientation estimates, which is essential for long-term SLAM performance.

All online metrics are computed post-hoc from exported trajectory files (`camera_poses.txt`), ensuring reproducibility and enabling detailed analysis without requiring access to the live reconstruction system.

Offline Reconstruction Metrics: For offline reconstruction, we employ COLMAP [2] to perform comprehensive structure-from-motion analysis. This choice allows us to evaluate reconstruction quality using established metrics in the photogrammetry community, providing a complementary perspective to the online DSO results. The offline evaluation encompasses both quantitative metrics and qualitative assessment:

Quantitative Metrics:

- *Reprojection Error:* The root mean square (RMS) reprojection error in pixels, computed across all registered images and 3D points. This metric directly measures the accuracy of the camera pose and 3D point estimates, with lower values indicating better geometric consistency.
- *Number of Reconstructed Points:* The total count of 3D points in the sparse reconstruction. This metric reflects the completeness of the reconstruction, with higher values generally indicating better scene coverage, though quality must be considered alongside quantity.
- *Registration Rate:* The percentage of input images successfully registered in the reconstruction, computed as $\frac{N_{\text{registered}}}{N_{\text{total}}} \times 100\%$. Higher registration rates indicate better robustness to challenging conditions such as illumination changes, motion blur, or insufficient texture.
- *Mean Observations per Image:* The average number of 3D points observed in each registered image, computed as $\frac{\sum_i N_{\text{obs},i}}{N_{\text{registered}}}$, where $N_{\text{obs},i}$ is the number of observations in image i . This metric reflects the density and quality of feature correspondences, with higher values indicating richer geometric constraints and potentially more robust reconstruction.

Qualitative Assessment: Beyond quantitative metrics, we perform visual inspection of both sparse and dense reconstructions to assess geometric accuracy, completeness, and overall reconstruction quality. The sparse reconstruction reveals the distribution and accuracy of feature correspondences, while the dense reconstruction (generated using COLMAP's MVS pipeline) provides insight into surface reconstruction quality and detail preservation. We particularly examine:

- *Sparse Reconstruction Quality:* Visual inspection of the 3D point cloud distribution, outlier presence, and geometric consistency of reconstructed structures.
- *Dense Reconstruction Quality:* Assessment of surface completeness, detail preservation, and absence of artifacts in the dense point cloud or mesh reconstruction.

4.4. Reconstruction Results

4.4.1. Online Reconstruction

Our online reconstruction experiments demonstrate the practical benefits of the photometric-stable pipeline in real-time scenarios. Table 2 presents a focused comparison between the raw and ours across the four key trajectory quality metrics on the *Corridor_I* dataset.

Table 2. Quantitative comparison of online reconstruction trajectory quality on the *Corridor_I* dataset: Raw vs. Ours

Metric	Raw	Ours
Translation Smoothness	0.9975	0.9959
Rotation Smoothness	0.9952	0.9964
Tracking Robustness	0.4991	0.6777
Temporal Consistency	0.7032	0.7801

The results reveal significant improvements in tracking stability metrics. Most notably, our method achieves substantially improved *tracking robustness* (0.6777 vs. 0.4991, a 35.8% relative improvement) and *temporal consistency* (0.7801 vs. 0.7032, a 10.9% improvement). These metrics directly reflect the stability of the visual odometry estimation, which is critical for reliable SLAM performance. The improvement in tracking robustness is particularly significant, as it indicates that our pipeline’s fixed gain exposure compensation and bilateral filtering contribute to more stable gradient computation, reducing the variance in motion estimates that can lead to tracking failures.

Our method also exhibits similar rotation smoothness (0.9964 vs. 0.9952) to raw one, which indicates our improvement retains the critical ability for applications requiring accurate orientation estimation. The translation smoothness values are comparable between the two paths (0.9975 vs. 0.9959), with both achieving excellent smoothness scores above 0.99, indicating that both approaches maintain high-quality translational tracking.

The substantial improvements in tracking robustness and temporal consistency validate our design philosophy of prioritizing photometric consistency over visual enhancement. These metrics directly translate to more reliable SLAM performance.

As illustrated in Figure 4, the qualitative visual comparison of the reconstructed trajectories provides compelling evidence for the superiority of our photometric-stable pipeline. The *corridor_I* sequence was captured with the camera moving along a predominantly linear path. The ground truth motion pattern should therefore manifest as a relatively straight trajectory in the reconstructed 3D space.

The raw path reconstruction (left panel) exhibits a pronounced initial curvature in the yellow trajectory, deviating significantly from the expected linear motion pattern. This initial deviation suggests that the raw processing pipeline suffers from systematic errors during the early stages of tracking,

likely attributable to photometric inconsistencies in the initial frames.

In contrast, our path reconstruction (right panel) demonstrates a substantially more accurate trajectory representation. The green trajectory closely follows a linear path, consistent with the actual camera motion during data acquisition. This improved accuracy can be directly attributed to our pipeline’s photometric stabilization mechanisms: (1) the fixed gain exposure compensation ensures consistent brightness levels across frames, preventing illumination-induced motion artifacts; (2) the gamma correction linearizes the image intensity response, enabling more accurate gradient computation for direct methods; and (3) the bilateral filtering preserves edge information while reducing noise, contributing to more stable feature tracking.

The trajectory accuracy improvement is particularly evident in the initial segment of the reconstruction, where the raw path shows a clear turning motion that does not correspond to the actual camera movement. This initial error can propagate through the entire reconstruction, as DSO’s direct method relies on frame-to-frame photometric consistency. Our pipeline’s ability to maintain accurate tracking from the outset demonstrates the critical importance of photometric stability for direct visual odometry methods.

The quantitative metrics corroborate these qualitative observations. The 35.8% improvement in tracking robustness directly reflects the reduced variance in motion estimates, which manifests visually as the more accurate, linear trajectory in our pipeline path. Similarly, the improved temporal consistency (10.9% increase) indicates that our pipeline maintains more stable orientation estimates, contributing to the overall trajectory accuracy.

Furthermore, the point cloud distributions in both reconstructions reveal subtle but important differences. While both paths successfully reconstruct the corridor structure, our path’s point cloud exhibits more consistent spatial distribution along the trajectory, with fewer outliers in regions corresponding to illumination transitions. This observation aligns with the improved tracking robustness metric and further validates the effectiveness of our photometric stabilization approach.

4.4.2. Offline Reconstruction

For offline reconstruction evaluation, we employ COLMAP [2] to perform comprehensive structure-from-motion analysis on pre-processed image sequences. This evaluation provides complementary insights to the online DSO results, leveraging COLMAP’s robust feature-based approach to assess reconstruction quality from a different perspective. We conduct experiments on three distinct datasets: an indoor object-centric scene (Psyduck) and two outdoor corridor sequences (*Corridor_L* and *Corridor_{Loop}*), each presenting unique challenges for visual SLAM systems.

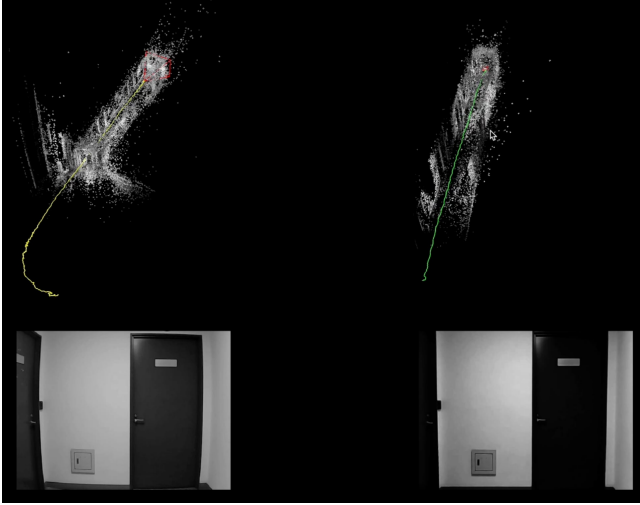


Fig. 4. Reconstruction Results of Raw (left) and Ours (right) using DSO

Table 3. Quantitative comparison of COLMAP reconstruction on the indoor Psyduck dataset.

Method	Reproj. Error	Reg. Rate	# Points (Sparse / Dense)	Mean Obs. / Image
Raw	1.168	100% (300)	8.1k / 1.08M	205.6
No Sharpen	1.091	95% (285)	13.7k / 0.95M	329.0
Proposed (Full)	1.062	100% (300)	18.0k / 1.00M	425.3
Standard (Full)	1.124	100% (300)	11.6k / 1.13M	304.1

Table 3 presents the COLMAP reconstruction results for the indoor Psyduck dataset, which serves as a baseline scenario with relatively stable lighting conditions.

The indoor Psyduck dataset results demonstrate clear advantages of our proposed pipeline over the baseline methods. Our method achieves the lowest reprojection error (1.06170 pixels), representing a 9.1% improvement over the raw path (1.16767 pixels) and a 5.5% improvement over the standard CLAHE approach (Standard, 1.12407 pixels). Notably, while the No.Sharpen variant achieves a competitive reprojection error (1.09086 pixels), it suffers from a reduced registration rate (95% vs. 100%), indicating that the sharpening step contributes to feature detection robustness in well-lit indoor environments.

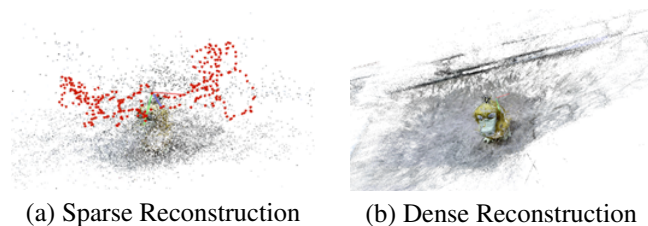


Fig. 5. COLMAP Reconstruction on Psyduck Dataset using Proposed (Full)

Table 4. Comparison of COLMAP reconstruction results on the challenging *Corridor_L* dataset.

Method	Reproj. Error	Reg. Rate	# Points (Sparse / Dense)	Mean Obs. / Image
Raw	0.944	41.8% (84)	4.1k / 0.18M	273.7
No Sharpen	1.042	42.3% (85)	3.9k / 0.15M	228.0
Proposed	1.011	41.8% (84)	4.3k / 0.15M	263.1
Standard	1.006	63.2% (127)	5.6k / 0.26M	247.0

The sparse point cloud statistics reveal that our proposed method produces the highest number of reconstructed 3D points (17,991), which is $2.2\times$ the raw path’s point count (8,073) and $1.6\times$ the Standard method (11,563). This substantial increase in point cloud density, combined with the highest mean observations per image (425.270), indicates that our custom CLAHE implementation provides superior feature detection and matching compared to OpenCV’s standard CLAHE. The higher observation count per image suggests richer geometric constraints, which directly contributes to more robust bundle adjustment and lower reprojection errors.

As shown in Figure 5, we can observe the qualitative results of sparse and dense reconstruction on Psyduck dataset with our method. The number of points of (b) is significantly more than that of (a), as illustrated.

Table 4 presents the results for the *Corridor_L* dataset, which features challenging illumination conditions with motion-activated sensor lights. In addition, *Corridor_L* covered a L-shape region, which is a critical geometrical prior for us to qualitatively assess the reconstruction result shown in Figure 6

The *Corridor_L* results reveal the significant challenges posed by abrupt illumination changes. Only the standard method produce a complete reconstruction, highlighting the critical importance of photometric preprocessing for direct methods in challenging lighting conditions. Among the processed methods, raw data as input achieves the lowest reprojection error (0.944 pixels), demonstrating superior geometric accuracy despite a lower registration rate (41.79%) compared to standard (63.2%). This trade-off suggests that while Standard’s aggressive CLAHE processing enables more images to be registered, it may introduce geometric distortions that manifest as higher reprojection errors.

The registration rate discrepancy is particularly noteworthy. Standard achieves a 51.2% relative improvement in registration rate (63.18% vs. 41.79%). This observation suggests that our custom CLAHE implementation provides a somehow too aggressive approach to balance between feature detectability and geometric consistency, resulting in slightly worse reconstruction completeness in this specific case of severe lighting variations. The higher mean observations per image for our method (263.143 vs. 246.965) further supports this interpretation, indicating that registered images in our pipeline contribute more reliable geometric constraints.

Table 5 presents the results for the *Corridor_{Loop}* dataset,

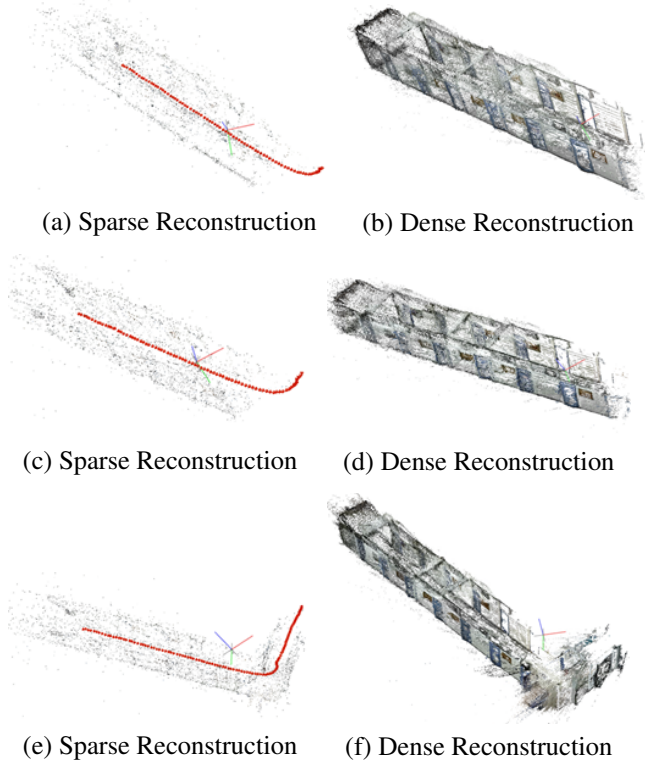


Fig. 6. COLMAP Reconstruction on *Corridor_L* Dataset using raw (a,b), ours (c,d), and OpenCV (e,f)

Table 5. Reconstruction results on the *Corridor_{Loop}* dataset. Note that while Standard (Full) achieves higher completeness, Proposed maintains a higher observation density.

Method	Reproj. Error	Reg. Rate	# Points (Sparse / Dense)	Mean Obs. / Image
Raw	0.964	28.1% (148)	6.0k / 0.27M	236.9
No Sharpen	1.073	28.8% (151)	6.1k / 0.24M	217.6
Proposed	1.045	26.3% (138)	6.5k / 0.23M	246.8
Standard	1.043	47.0% (246)	8.0k / 0.40M	187.1

which represents a longer sequence with more complex geometry and extremely severe lighting variations. The scene is captured in a loop-like corridors in the shape of a square. Thus, the complete reconstruction should be connected as a square.

The *Corridor_{Loop}* dataset results present an interesting contrast. The raw path achieves the lowest reprojection error (0.963969 pixels) but suffers from a severely limited registration rate (28.14%), indicating that while the raw images provide accurate geometric information when successfully matched, the lack of preprocessing severely restricts the number of images that can be registered. This low registration rate is a critical limitation for practical SLAM or even offline 3D reconstruction applications, as it results in incomplete map coverage.

Our method maintains competitive reprojection accuracy (1.04477 pixels) while achieving a similar registration rate to the raw path (26.34% vs. 28.14%). However, Standard demonstrates a substantial improvement in registration rate (46.95%), nearly doubling the coverage compared to both raw and Paper methods. The differences between these reconstructions with different registration rate can be clearly demonstrated in Figure 7 where only the reconstruction utilizing Standard method can reconstruct the second edge of a square-like corridor of a loop. This suggests that in extremely challenging outdoor scenarios with long sequences, OpenCV’s standard CLAHE may provide more aggressive contrast enhancement that enables feature detection in a wider variety of lighting conditions, albeit at the cost of slightly higher reprojection errors.

The point cloud statistics reveal that Standard produces the densest sparse reconstruction (7,968 points) and the most comprehensive dense point cloud (399,427 points), which aligns with its superior registration rate. However, the mean observations per image metric (187.081) is lower than both raw (236.946) and ours (246.768), suggesting that while Standard registers more images, the geometric constraints per image may be less reliable, potentially explaining the marginally higher reprojection error.

4.5. Ablation Study on Feature Matching

To validate the efficacy of individual components in our pre-processing pipeline, we conduct an ablation study focusing on the fundamental unit of SfM: feature matching. We utilize SIFT feature extraction and exhaustive matching on representative image pairs to quantify the impact of photometric enhancement.

Impact of Sharpening in Stable Environments: Figure 8 visualizes the feature matching performance on the indoor *Psyduck* dataset. The results demonstrate a progressive improvement in match density. The raw image pair (a) yields only 171 matches. The standard OpenCV CLAHE (c) improves this to 281 matches. However, our proposed full pipeline (b), which integrates custom CLAHE with unsharp masking, achieves the highest number of inlier matches (**386**). This represents a **125% increase** over the raw input and a **37% increase** over the standard OpenCV approach. The significant jump from standard CLAHE to our method highlights the contribution of the sharpening step; by enhancing high-frequency details on the texture-less surfaces of the object, our pipeline enables the SIFT detector to identify stable keypoints that are otherwise invisible.

Performance under Varying Illumination: Figure 9 examines the *Corridor_L* dataset, where sensor lights create inconsistent illumination.

1. **Raw vs. Enhanced:** The raw images (a) suffer from low contrast, resulting in the fewest matches (101).

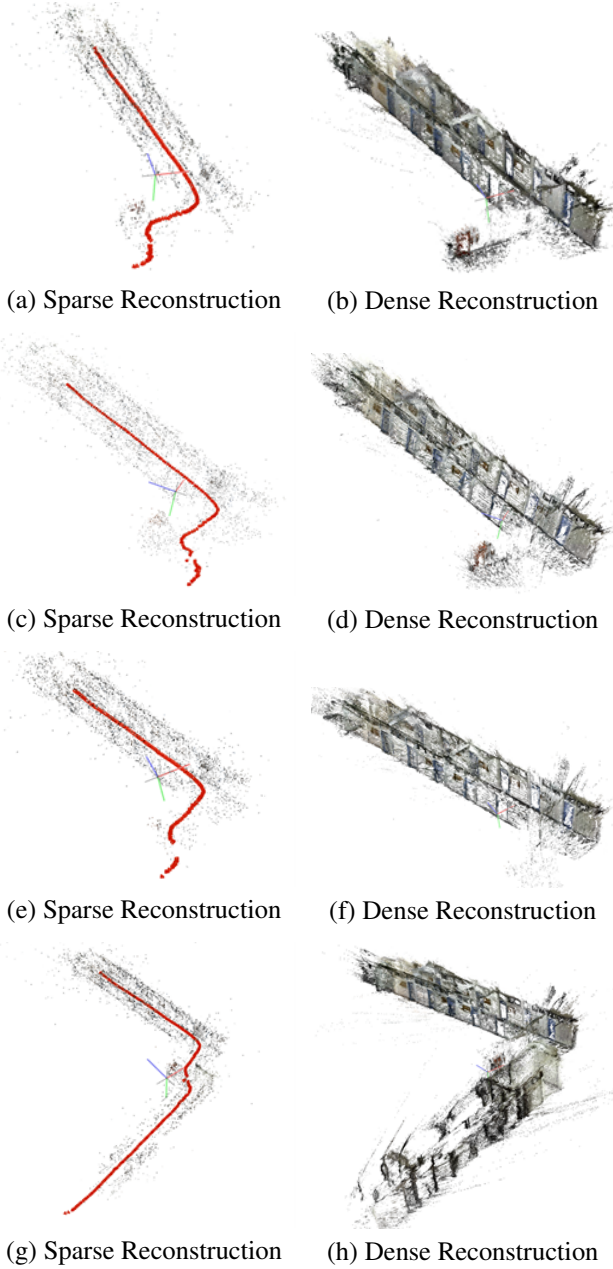


Fig. 7. COLMAP Reconstruction on *Corridor_{Loop}* Dataset using raw (a,b), no_sharpen (c,d), ours (e,f), and OpenCV (g,h). Significant improvement can be observed in (g) and (h) where the reconstruction is more complete and the registration rate of images is higher than that of others

2. **Effect of Sharpening:** Comparing the *No_Sharpen* variant (b) with the *Proposed* full pipeline (c) reveals that sharpening contributes an additional gain (144 → **157** matches). Even in noisy, low-light environments, the sharpening filter successfully amplifies structural edges without introducing fatal noise artifacts.
3. **Comparison with Standard:** Notably, on this specific image pair, our method outperforms the OpenCV standard implementation (157 vs. 136 matches). This indicates that our custom CLAHE parameterization preserves local feature distinctiveness better than the default adaptive histogram equalization, which often over-amplifies noise in flat regions.

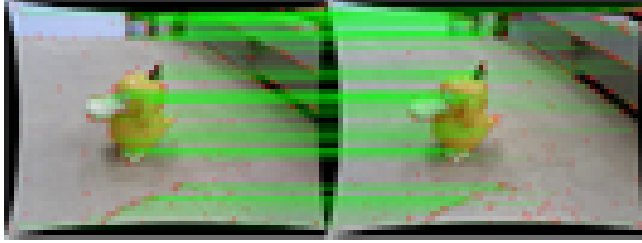
Discussion on Precision-Recall Trade-off: A critical analysis is required to reconcile the results in Figure 9 with Table 4. While Figure 9 shows that our method generates *more matches* on successful pairs (157 vs. 136), Table 4 previously indicated that the Standard method achieves a higher *global registration rate* (63.2% vs. 41.8%).

This discrepancy suggests a distinct trade-off between **reconstruction precision** and **registration recall**. Our pipeline is designed to maximize geometric consistency (evidenced by the higher match count per pair and lower reprojection error), but this strictness makes the feature matcher more conservative. In contrast, the Standard OpenCV method appears to be more aggressive, allowing it to register marginally relatable frames (higher recall) at the cost of lower match density per pair and potentially higher geometric error. Thus, while our method provides higher fidelity for the parts of the scene it reconstructs, the Standard method exhibits higher tolerance for extreme illumination gaps in long sequences.

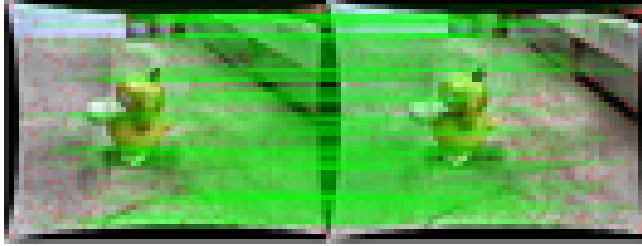
5. CONCLUSION

In this work, we presented a comprehensive framework to enhance the performance of visual SLAM and Structure-from-Motion (SfM) using commodity USB cameras. By acknowledging the distinct requirements of direct methods and feature-based approaches, we designed two specialized pre-processing pipelines that effectively mitigate the limitations of low-cost sensors, such as high-frequency noise, lens distortion, and auto-exposure instability.

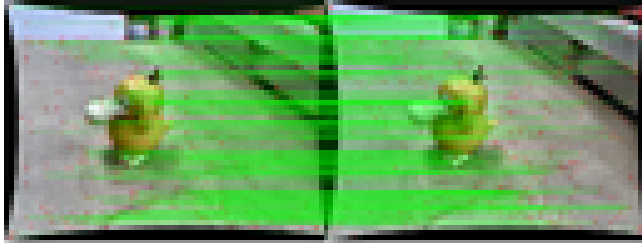
For online real-time reconstruction, our results demonstrate that photometric stability is paramount. The proposed online pipeline, characterized by fixed-gain exposure compensation and lightweight bilateral filtering, successfully stabilized the photometric error surface required by DSO. Quantitative evaluations on the *Corridor* datasets revealed a significant 35.8% improvement in tracking robustness and a 10.9% increase in temporal consistency compared to raw input. Qualitatively, this led to the elimination of spurious curvature in trajectory estimation, proving that suppressing



(a) Raw: 171 Matches



(b) Our: 386 Matches



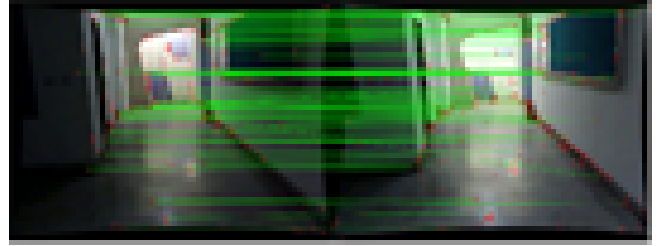
(c) OpenCV: 281 Matches

Fig. 8. Ablation Study of Feature Matching on *Psyduck* dataset

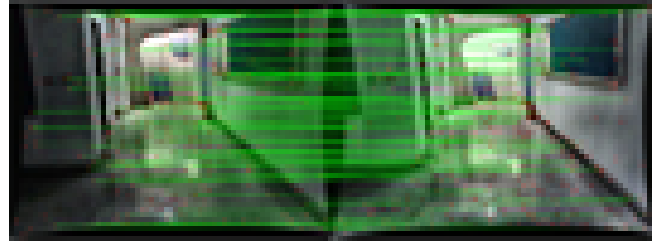
temporal brightness fluctuations is more critical for direct VO than maximizing local image contrast.

Conversely, our offline pipeline prioritized feature recoverability for SfM applications. Through the implementation of adaptive Tuned-CLAHE and unsharp masking, we maximized the extraction of SIFT keypoints in texture-poor environments. In the indoor *Psyduck* dataset, our method achieved a $2.2\times$ increase in sparse point cloud density and superior reprojection accuracy compared to raw and standard processing methods. However, experiments in challenging outdoor scenarios with extreme lighting variations (*Corridor_L*) highlighted a trade-off; while our method consistently yielded high-quality feature observations with low reprojection errors, standard aggressive contrast enhancement sometimes achieved higher registration rates at the expense of geometric precision.

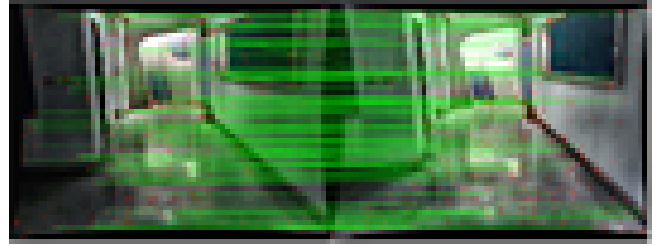
Collectively, these findings underscore that there is no "one-size-fits-all" image processing solution for 3D reconstruction. Direct methods demand global photometric consistency to minimize linearization errors, whereas feature-based methods benefit from local contrast enhancement to



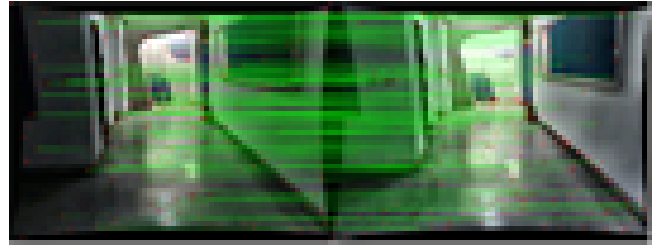
(a) Raw: 101 Matches



(b) No_Sharpen: 144 Matches



(c) Our: 157 Matches



(d) OpenCV: 136 Matches

Fig. 9. Ablation Study of Feature Matching on *Corridor_L* dataset. Our method (c) has the most number of matched features using COLMAP feature matching module, where both (b), (c), and (d) all outperform raw (a) significantly. With more matched features, it is more possible for the reconstruction to complete the overall structure when the scene is difficult to reconstruct (e.g. severe lighting variations or texture-less surfaces)

maximize descriptor distinctiveness. By tailoring the signal processing pipeline to the specific mathematical assumptions of the downstream algorithm, DSO and COLMAP, we successfully bridged the gap between low-cost hardware and high-precision 3D reconstruction. Future work may explore learning-based exposure control to dynamically balance these trade-offs in environments with high dynamic range.

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